

Setting up Dataiku on Azure

by Mohamad Ikhsan Nurulloh



Disclaimer

- **No Endorsement:** This guide is for educational purposes only and does not imply endorsement by Dataiku or Microsoft Azure.
- **Testing Only:** Intended for testing and development, not production environments.
- **Refer to Official Docs:** Always consult official Dataiku and Azure documentation for production use.
- **Proceed at Your Own Risk:** Use this guide at your own risk and be aware of potential deployment implications.



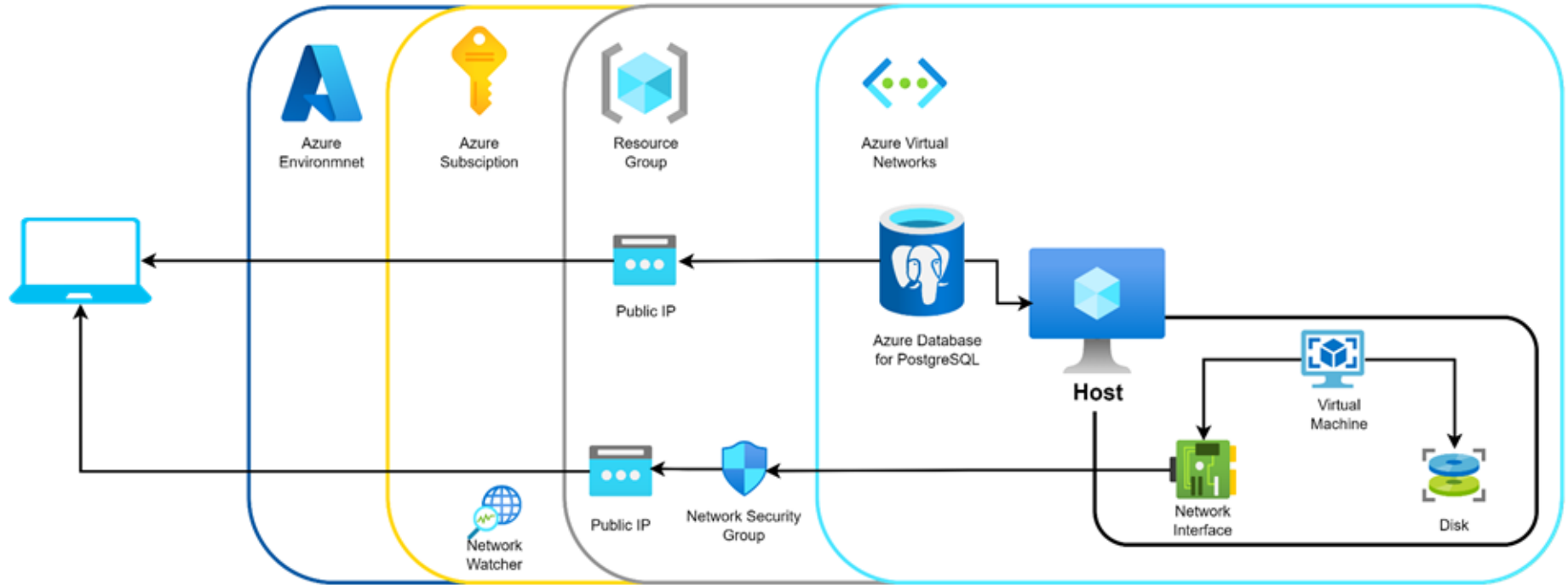
data iku

Dataiku is a platform that democratizes data and AI, helping organizations drive business value. It offers data integration, preparation, and development capabilities, integrating with various infrastructures and supporting model summaries using generative AI. Dataiku is committed to diversity and inclusivity, with initiatives aimed at using AI for good.

Before Installing Dataiku on Azure

- **Check System Requirements:** Verify Azure environment meets Dataiku prerequisites.
- **Consult Documentation:** Review official Dataiku and Azure installation guides.
- **Plan for Security:** Implement data protection and access controls.
- **Consider Costs:** Understand Azure and Dataiku licensing expenses.
- **Backup and Recovery:** Prepare a backup and recovery plan.
- **Testing Environment:** Validate installation in a testing setup first.

Dataiku on Azure Architecture



Dataiku on Azure Architecture Explanation

- **Azure Platform:** Dataiku is hosted on Azure, requiring an Azure subscription for access and management. Resources are organized within resource groups.
- **Networking and Security:** Public IPs enable external access, while Network Security Groups (NSGs) and Network Watcher manage and secure network traffic.
- **Virtual Networking:** Azure Virtual Networks (VNETs) create isolated networks for resource communication and connectivity.
- **Database and Compute:** Azure Database for PostgreSQL provides scalable, managed database services, and Virtual Machines (VMs) run Dataiku applications. Disks offer storage for VMs.

Summary Azure Resource

Microsoft Azure Upgrade Search resources, services, and docs (G+)

All services > All resources Default Directory

+ Create Manage view Refresh Export to CSV Open query Assign tags Delete

Filter for any field... Subscription equals all Resource group equals all Type equals all Location equals all Add filter

2 Recommendations 0 Changed resources 3 Unsecure resources No grouping List view

☐ Name ↑↓	Type ↑↓	Resource group ↑↓	Location ↑↓	Subscription ↑↓	
☐ dataiku	Virtual machine	Ikhsan	Korea Central	Azure subscription 1	...
☐ dataiku	Virtual network	Ikhsan	Korea Central	Azure subscription 1	...
☐ dataiku-ip	Public IP address	Ikhsan	Korea Central	Azure subscription 1	...
☐ dataiku-nsg	Network security group	Ikhsan	Korea Central	Azure subscription 1	...
☐ dataiku540	Network Interface	Ikhsan	Korea Central	Azure subscription 1	...
☐ dataiku_OsDisk_1_e489aa3edf684f45a73c3f9bd3be468f	Disk	IKHSAN	Korea Central	Azure subscription 1	...
☐ dataikudataset	Azure Database for PostgreSQL flexible ...	Ikhsan	Korea Central	Azure subscription 1	...
☐ NetworkWatcher_koreacentral	Network Watcher	NetworkWatcherRG	Korea Central	Azure subscription 1	...

The Azure resources used for the installation and configuration of Dataiku.

Creating Virtual Network

1. Defining Projects and Instance Details

Microsoft Azure

Upgrade

Search resources, services, and docs (G+J)

ikhsannur1996@outloo...
DEFAULT DIRECTORY (IKH-SANNUR...)

All services > Virtual network >

Create virtual network

BasicsSecurityIP addressesTagsReview + create

Project details

Select the subscription to manage deployed resources and costs. Use resource groups like folders to organize and manage all your resources.

Subscription *

Azure subscription 1

Resource group *

ikhsan

Create new

Instance details

Virtual network name *

dataiku

Region * ⓘ

(Asia Pacific) Korea Central

Deploy to an Azure Extended Zone

Previous

Next

Review + create

Give feedback

Define the projects and instance details for your virtual network setup.

2. Enhancing Security by Subscribing to Security Services

Microsoft Azure

Upgrade

Search resources, services, and docs (G+)



ikhsannur1996@outloo...
DEFAULT DIRECTORY (IKHSANNUR...)

[All services](#) > [Virtual network](#) >

Create virtual network

Basics **Security** IP addresses Tags Review + create

Enhance the security of your virtual network with these additional paid security services. [Learn more](#)

Virtual network encryption

Enable Virtual network encryption to encrypt traffic traveling within the virtual network. Virtual machines must have accelerated networking enabled. Traffic to public IP addresses is not encrypted. [Learn more](#)

Virtual network encryption ☐

Azure Bastion

Azure Bastion is a paid service that provides secure RDP/SSH connectivity to your virtual machines over TLS. When you connect via Azure Bastion, your virtual machines do not need a public IP address. [Learn more](#)

Enable Azure Bastion ☐

Previous

Next

Review + create

Give feedback

Enhance the security of your virtual network by subscribing to Azure security services.

3. Defining IP address Range

Microsoft Azure

Upgrade

Search resources, services, and docs (G+)



ikhsannur1996@outloo...
DEFAULT DIRECTORY (IKHSANNU...)

All services > Virtual network >

Create virtual network

BasicsSecurityIP addressesTagsReview + create

Define the address space of your virtual network with one or more IPv4 or IPv6 address ranges. Create subnets to segment the virtual network address space into smaller ranges for use by your applications. When you deploy resources into a subnet, Azure assigns the resource an IP address from the subnet. [Learn more](#)

Add IPv4 address space

10.0.0.0/16

Delete address space

10.0.0.0

/16

10.0.0.0 - 10.0.255.25565,536 addresses

+ Add a subnet

Subnets	IP address range	Size	NAT gateway
default	10.0.0.0 - 10.0.255	/24 (256 addresses)	-

Previous

Next

Review + create

Give feedback

Define the IP address range for your virtual network.

4 Add tags

Microsoft Azure

Upgrade

Search resources, services, and docs (G+J)



ikhsmnur1996@outloo...
DEFAULT DIRECTORY (IKHSMN...

All services > Virtual network >

Create virtual network ...

Basics Security IP addresses **Tags** Review + create

Tags are name/value pairs that enable you to categorize resources and view consolidated billing by applying the same tag to multiple resources and resource groups. [Learn more about tags](#)

Note that if you create tags and then change resource settings on other tabs, your tags will be automatically updated.

Name	Value	Resource
project	dataiku	All resources selected
		All resources selected

Previous

Next

Review + create

Give feedback

Add tags to your virtual network for better organization and management.

5. Review and Create

Microsoft Azure

Upgrade

Search resources, services, and docs (G+/I)



ikhsannur1996@outloo...
DEFAULT DIRECTORY (IKHSANN...)

All services > Virtual network >

Create virtual network ...

BasicsSecurityIP addressesTagsReview + create

[View automation template](#)

Basics

Subscription	Azure subscription 1
Resource Group	Ikhsan
Name	dataiku
Region	Korea Central

Security

Azure Bastion	Disabled
Azure Firewall	Disabled
Azure DDoS Network Protection	Disabled

IP addresses

Address space	10.0.0.0/16 (65,536 addresses)
---------------	--------------------------------

PreviousNextCreate

Give feedback

Review all the details and create the virtual network.

5. Network Success Created

The screenshot displays the Microsoft Azure portal interface. At the top, the header shows 'Microsoft Azure' with an 'Upgrade' button, a search bar, and user information for 'ikhsannur1996@outloo...'. The main content area is titled 'dataiku' (Virtual network) and includes a search bar and navigation links like 'Move', 'Delete', 'Refresh', and 'Give feedback'. A left-hand navigation pane lists various settings such as 'Activity log', 'Access control (IAM)', 'Tags', 'Diagnose and solve problems', 'Settings', 'Address space', 'Connected devices', 'Subnets', 'Bastion', 'DDoS protection', 'Firewall', 'Microsoft Defender for Cloud', 'Network manager', 'DNS servers', and 'Peerings'. The main panel shows the 'Overview' tab for the 'dataiku' virtual network. It lists essential details: Resource group (ikhsan), Location (Korea Central), Subscription (Azure_subscription_1), and Subscription ID. It also shows the Address space (10.0.0.0/16), DNS servers (Azure provided DNS service), Flow timeout (Configure), BGP community string (Configure), and Virtual network ID (01a07f11-741e-4b40-87f6-57c12d35eb7d). Below this, there are tabs for 'Topology', 'Properties', 'Capabilities (5)', 'Recommendations', and 'Tutorials'. The 'Capabilities (5)' tab is active, showing four cards: 'DDoS protection' (Not configured), 'Azure Firewall' (Not configured), 'Peerings' (Not configured), and 'Microsoft Defender for Cloud' (Strengthen the security posture of your environment.). A fifth card, 'Private endpoints', is partially visible at the bottom.

Microsoft Azure Upgrade Search resources, services, and docs (G+/)

All services > dataiku-1722042051313 | Overview >

dataiku Virtual network

Search

Move Delete Refresh Give feedback

JSON View

Essentials

Resource group (move)	: ikhsan	Address space	: 10.0.0.0/16
Location (move)	: Korea Central	DNS servers	: Azure provided DNS service
Subscription (move)	: Azure_subscription_1	Flow timeout	: Configure
Subscription ID	: 5a3b05d8-356e-4889-9cdd-c85226ae54f2	BGP community string	: Configure
		Virtual network ID	: 01a07f11-741e-4b40-87f6-57c12d35eb7d

Tags (edit) : project : dataiku

Topology Properties **Capabilities (5)** Recommendations Tutorials

DDoS protection

Configure additional protection from distributed denial of service attacks.

● Not configured

Azure Firewall

Protect your network with a stateful L3-L7 firewall.

● Not configured

Peerings

Seamlessly connect two or more virtual networks.

● Not configured

Microsoft Defender for Cloud

Strengthen the security posture of your environment.

Private endpoints

Confirm that the virtual network has been successfully created.

6. Configure Custom Inbound Rule

The screenshot shows the Microsoft Azure portal interface. The top navigation bar includes the Microsoft Azure logo, an 'Upgrade' button, a search bar, and user information. The left sidebar shows the 'Network settings' menu item selected under the 'Networking' section. The main content area displays the 'Network security group dataiku-nsg' configuration page. The page title is 'dataiku | Network settings' with a star icon. Below the title, there's a search bar and a message: 'This is a new experience. Please provide feedback'. The 'Rules' section is expanded, showing a list of inbound port rules. The rules are filtered by 'Source == all', 'Destination == all', 'Protocol == all', and 'Action == all'. The table lists 7 inbound port rules, including SSH, HTTP, HTTPS, and a custom rule 'AllowAnyCustom11000Inbound'. The custom rule is highlighted in blue. Below the table, there's a link to 'Outbound port rules (3)'.

Microsoft Azure Upgrade Search resources, services, and docs (G+/I) ikhsannur1996@outloo... DEFAULT DIRECTORY (IKHSANNUR...)

All services > dataiku

dataiku | Network settings ☆ ... Virtual machine

Search

This is a new experience. Please provide feedback

Rules Collapse all

Network security group dataiku-nsg (attached to networkinterface: dataiku540) Impacts 0 subnets, 1 network interfaces + Create port rule

Search rules Source == all Destination == all Protocol == all Action == all

Priority ↑	Name	Port	Protocol	Source	Destination	Action
Inbound port rules (7)						
300	SSH	22	TCP	Any	Any	Allow
320	HTTP	80	TCP	Any	Any	Allow
340	HTTPS	443	TCP	Any	Any	Allow
350	AllowAnyCustom11000Inbound	11000	Any	Any	Any	Allow
65000	AllowVnetInbound	Any	Any	VirtualNetwork	VirtualNetwork	Allow
65001	AllowAzureLoadBalancerInbound	Any	Any	AzureLoadBalancer	Any	Allow
65500	DenyAllInbound	Any	Any	Any	Any	Deny
Outbound port rules (3)						

Add a custom inbound rule to allow traffic on the Dataiku port.

Creating Azure Database for PostgreSQL

1. Creating PostgreSQL Projects Details

Microsoft Azure

Upgrade

Search resources, services, and docs (G+/)



ikhannur1996@outloo...
DEFAULT DIRECTORY (IKHANNUR1996@OUTLOO...)

All services > Azure Database for PostgreSQL servers >

New Azure Database for PostgreSQL Flexible server

Microsoft

Server names, networking connectivity method and backup redundancy cannot be changed after server is created. Review these options carefully before provisioning.

Project details

Select the subscription to manage deployed resources and costs. Use resource groups like folders to organize and manage all your resources.

Subscription *

Azure subscription 1

Resource group *

Ikhsan

Create new

Server details

Enter required settings for this server, including picking a location and configuring the compute and storage resources.

Server name *

dataikudataset

Region *

Korea Central

PostgreSQL version *

16

Workload type

☒ Development

☐ Production

Estimated costs

Compute Sku	USD 18.98/month
Free upto 750 hours	
Standard_B1ms (1 vCore)	18.98
Storage	USD 4.19/month
Free upto 32 GB	
Storage selected 32 GiB (USD 0.13 per GiB)	32 x 0.13
Bandwidth	
For outbound data transfer across services in different regions will incur additional charges. Any inbound data transfer is free. Learn more	
Estimated total	USD 23.17/month

Review + create

Next : Networking >

Define the projects and server details for your PostgreSQL

2. Configuring Network

The screenshot displays the 'New Azure Database for PostgreSQL Flexible server' configuration page in the Microsoft Azure portal. The 'Networking' tab is selected, showing options for network connectivity and estimated costs.

Microsoft Azure Upgrade Search resources, services, and docs (G+)

All services > Azure Database for PostgreSQL servers >

New Azure Database for PostgreSQL Flexible server

Microsoft

⚠ Server names, networking connectivity method and backup redundancy cannot be changed after server is created. Review these options carefully before provisioning.

Basics **Networking** Security Tags Review + create

Configure networking access and security for your server.

Network connectivity

You can connect to your server by specifying a public IP address, creating private endpoints or from within a selected virtual network.

Connectivity method ⓘ

- ☒ Public access (allowed IP addresses) and Private endpoint
- ☐ Private access (VNet Integration)

📘 Connections from the IP addresses configured in the Firewall rules section below will have access to this server. By default, no public IP addresses are allowed. [Learn more](#)

Public access

☒ Allow public access to this resource through the internet using a public IP address ⓘ

Estimated costs

Compute Sku	USD 18.98/month
Free upto 750 hours	
Standard_B1ms (1 vCore)	18.98
Storage	USD 4.19/month
Free upto 32 GB	
Storage selected 32 GiB (USD 0.13 per GiB)	32 x 0.13
Bandwidth	
For outbound data transfer across services in different regions will incur additional charges. Any inbound data transfer is free. Learn more	
Estimated total	USD 23.17/month

[Review + create](#) < Previous Next : Security >

Configure the network settings for your PostgreSQL database.

3. Configure Database Security

Microsoft Azure

Upgrade

Search resources, services, and docs (G+)

ikhsannur1996@outloo...
DEFAULT DIRECTORY (IKHSANNUR...)

All services > Azure Database for PostgreSQL servers >

New Azure Database for PostgreSQL Flexible server

Microsoft

⚠ Server names, networking connectivity method and backup redundancy cannot be changed after server is created. Review these options carefully before provisioning.

Basics Networking **Security** Tags Review + create

Configure data security features for your server.

Data encryption

The storage used for database and backup is encrypted by default with service managed keys. Data Encryption additionally encrypts your databases, backups, and logs at rest without any changes to your application. To use a customer managed key for data encryption, you'll need to select a managed identity and a key to which the selected identity has access. [Learn more](#)

Data encryption key

☒ Service-managed key

☐ Customer-managed key

Estimated costs

Compute Sku	USD 18.98/month
Free upto 750 hours	
Standard_B1ms (1 vCore)	18.98
Storage	USD 4.19/month
Free upto 32 GB	
Storage selected 32 GiB (USD 0.13 per GiB)	32 x 0.13
Bandwidth	
For outbound data transfer across services in different regions will incur additional charges. Any inbound data transfer is free. Learn more	
Estimated total	USD 23.17/month

Review + create

< Previous

Next : Tags >

Configure the security settings for your PostgreSQL database.

4. Add Tags

Microsoft Azure Upgrade Search resources, services, and docs (G+/)

All services > Azure Database for PostgreSQL servers >

New Azure Database for PostgreSQL Flexible server

Microsoft

⚠ Server names, networking connectivity method and backup redundancy cannot be changed after server is created. Review these options carefully before provisioning.

Basics Networking Security **Tags** Review + create

Tags are name/value pairs that enable you to categorize and view consolidated billing by applying the same tag to multiple resources and resource groups. [Learn more](#)

Note that if you create tags and then change resource settings on other tabs, your tags will be automatically updated.

Name	Value	Resource
project	dataiku	Server
		Server

Estimated costs

- Compute Sku** USD 18.98/month
 - Free upto 750 hours
 - Standard_B1ms (1 vCore) 18.98
- Storage** USD 4.19/month
 - Free upto 32 GB
 - Storage selected 32 GiB (USD 0.13 per GiB) 32 x 0.13
- Bandwidth**

For outbound data transfer across services in different regions will incur additional charges. Any inbound data transfer is free. [Learn more](#)

Estimated total **USD 23.17/month**

[Review + create](#) < Previous Next : Review + create >

Add tags to your PostgreSQL database for better organization and management.

5. Review + Create Azure Database for PostgreSQL

The screenshot shows the 'Review + create' step in the Azure portal for a new PostgreSQL Flexible server. The interface includes a top navigation bar with the Microsoft Azure logo, an 'Upgrade' button, a search bar, and user information. The main content area is divided into two sections: 'Product details' and 'Basics (Change)'. The 'Basics (Change)' section contains a table with configuration details. To the right, a 'Summary' box displays the estimated costs for the configuration.

Product details

Azure Database for PostgreSQL
by Microsoft
[Terms of use](#) | [Privacy policy](#)

Basics (Change)

Configuration	Value
Subscription	Azure subscription 1
Resource group	ikhsan
Server name	dataikudataset
Server admin login name	postgres
Location	Korea Central
Availability zone	No preference
High availability	Not Enabled
PostgreSQL version	16
Compute + storage	Burstable, 81ms, 1 vCores, 2 GiB RAM, 32 GiB storage, P4 (120 IOPS)
Backup retention period (in days)	7 day(s)

Estimated costs

Category	Cost
Compute Sku	USD 18.98/month
Free upto 750 hours	
Standard_B1ms (1 vCore)	18.98
Storage	USD 4.19/month
Free upto 32 GiB	
Storage selected 32 GiB (USD 0.13 per GiB)	32 x 0.13
Bandwidth	
For outbound data transfer across services in different regions will incur additional charges. Any inbound data transfer is free. Learn more	
Estimated total	USD 23.17/month

Charges will apply if you use above the free monthly limits. Please check [your usage](#)

[Create](#) [Previous](#) [Download a template for automation](#)

Review all the details and create the PostgreSQL database.

6. Success Create Azure Database for PostgreSQL

Microsoft Azure

Upgrade

Search resources, services, and docs (G+ /)



ikhsannur1996@outloo...
DEFAULT DIRECTORY (IKHSANNUR...)

Home > PostgreSQLFlexibleServer_01257059556c438399826aea99ea0ae6 | Overview >

 **dataikudataset** ☆ ☆ ...
Azure Database for PostgreSQL flexible server

Search



Overview

Activity log

Access control (IAM)

Tags

Diagnose and solve problems

Migration

Settings

Power Platform

Security

Intelligent Performance

Monitoring

Automation

Help

Essentials

Subscription (move) : [Azure subscription 1](#)

Subscription ID : 5a3b05d8-356e-4889-9cdd-c85226ae54f2

Resource group (move) : [ikhsan](#)

Status : Available

Location : Korea Central

Tags (edit) : project : dataiku

Server name : dataikudataset.postgres.database.azure.com

Server admin login name : postgres

Configuration : [Burstable_B1ms_1_vCores_2_GiB_RAM_32_GiB_storage](#)

PostgreSQL version : 16.3

Availability zone : 3

High availability : Not Enabled

Created On : 2024-07-27 01:29:12.7123006 UTC

JSON View

Getting started

Properties

Recommendations

Monitoring

Tutorials

Start your project

Connect to your database for the first time with a few simple steps.

**Allow access**
Configure network access to your PostgreSQL database

**Connect**
View connection string to learn how to connect with the application driver you use (.NET, PSQL, Python, JDBC, PHP, Node.js, Ruby, PHP, C++)

**Samples**
Setup a sample database schema to get started

Confirm that the PostgreSQL database has been successfully created.

Creating Virtual Machine

1. Define Virtual Machine Project

Microsoft Azure Upgrade Search resources, services, and docs (G+ /)

All services > Virtual machines >

Create a virtual machine

Project details

Select the subscription to manage deployed resources and costs. Use resource groups like folders to organize and manage all your resources.

Subscription * Azure subscription 1

Resource group * Ikhsan

[Create new](#)

Instance details

Virtual machine name * dataiku

Region * (Asia Pacific) Korea Central

Availability options No infrastructure redundancy required

Security type Trusted launch virtual machines

[Configure security features](#)

Image * Debian 11 "Bullseye" - x64 Gen2 (free services eligible)

[See all images](#) | [Configure VM generation](#)

< Previous Next : Disks > Review + create

[Give feedback](#)

Set up a virtual machine with Debian 11 as the operating system to run and manage Dataiku.

2. Choose Virtual Machine Authentication Method

Microsoft Azure Upgrade Search resources, services, and docs (G+)

All services > Virtual machines >

Create a virtual machine

Size * See all sizes

Enable Hibernation ☐

Administrator account

Authentication type ☐ SSH public key ☒ Password

Username *

Password *

Confirm password *

Inbound port rules

Select which virtual machine network ports are accessible from the public internet. You can specify more limited or granular network access on the Networking tab.

< Previous Next : Disks > Review + create

Give feedback

Configure SSH access to the VM using a password for straightforward login and management, ensuring the password is strong and secure to maintain system security.

3. Define Inbound Rules

Microsoft Azure Upgrade Search resources, services, and docs (G+)

All services > Virtual machines >

Create a virtual machine

Authentication type

☐ SSH public key

☒ Password

Username * ikhsannur1996 ✓

Password * ✓

Confirm password * ✓

Inbound port rules

Select which virtual machine network ports are accessible from the public internet. You can specify more limited or granular network access on the Networking tab.

Public inbound ports *

☐ None

☒ Allow selected ports

Select inbound ports * HTTP (80), HTTPS (443), SSH (22) ✓

i All traffic from the internet will be blocked by default. You will be able to change inbound port rules in the VM > Networking page.

< Previous Next : Disks > Review + create

Give feedback

Configure the VM's firewall to allow incoming connections on ports 80 (HTTP), 443 (HTTPS), and 22 (SSH) to enable web access and remote management.

4. Configuring Virtual Machine Disk

Microsoft Azure Upgrade Search resources, services, and docs (G+J)

All services > Virtual machines >

Create a virtual machine

OS disk

OS disk size 128 GiB (P10)

Some images are, by default, smaller than the selected OS disk size. [Click here to learn how to expand your disk partition size after you create your VM.](#)

OS disk type * Premium SSD (locally-redundant storage)

Delete with VM ☒

Key management Platform-managed key

Enable Ultra Disk compatibility ☐

Ultra disk is not supported for the selected VM size Standard_E4-2as_v5 in Korea Central.

Data disks for dataiku

You can add and configure additional data disks for your virtual machine or attach existing disks. This VM also comes with a temporary disk.

< Previous Next: Networking > Review + create

Give feedback

Opt for an SSD as the primary storage for the VM to enhance performance with faster data read/write speeds, which benefits Dataiku and overall system responsiveness.

5. Select Virtual Network

Microsoft Azure

Upgrade

Search resources, services, and docs (G+)



ikh sannur1996@outloo...
DEFAULT DIRECTORY (IKH-SANNUR...)

All services > Virtual machines >

Create a virtual machine

BasicsDisksNetworkingManagementMonitoringAdvancedTagsReview + create

Define network connectivity for your virtual machine by configuring network interface card (NIC) settings. You can control ports, inbound and outbound connectivity with security group rules, or place behind an existing load balancing solution.
[Learn more](#)

Network interface

When creating a virtual machine, a network interface will be created for you.

Virtual network *

dataiku

Create new

Subnet *

default (10.0.0.0/24)

Manage subnet configuration

Public IP

(new) dataiku-ip

Create new

NIC network security group

☐ None

☒ Basic

☐ Advanced

Public inbound ports *

☐ None

< Previous

Next : Management >

Review + create

 Give feedback

Select the virtual network for your virtual machines.

6. Select Inbound Network

Microsoft Azure

Upgrade

Search resources, services, and docs (G+)

ikhsannur1996@outloo...
DEFAULT DIRECTORY (IKHSANN...)

[All services](#) > [Virtual machines](#) >

Create a virtual machine

Public inbound ports *

☐ None

☒ Allow selected ports

Select inbound ports *

HTTP (80), HTTPS (443), SSH (22)

This will allow all IP addresses to access your virtual machine. This is only recommended for testing. Use the Advanced controls in the Networking tab to create rules to limit inbound traffic to known IP addresses.

Delete public IP and NIC when VM is deleted

☐

Enable accelerated networking

☒

Load balancing

You can place this virtual machine in the backend pool of an existing Azure load balancing solution. [Learn more](#)

Load balancing options

☒ None

☐ Azure load balancer
Supports all TCP/UDP network traffic, port-forwarding, and outbound flows.

< Previous

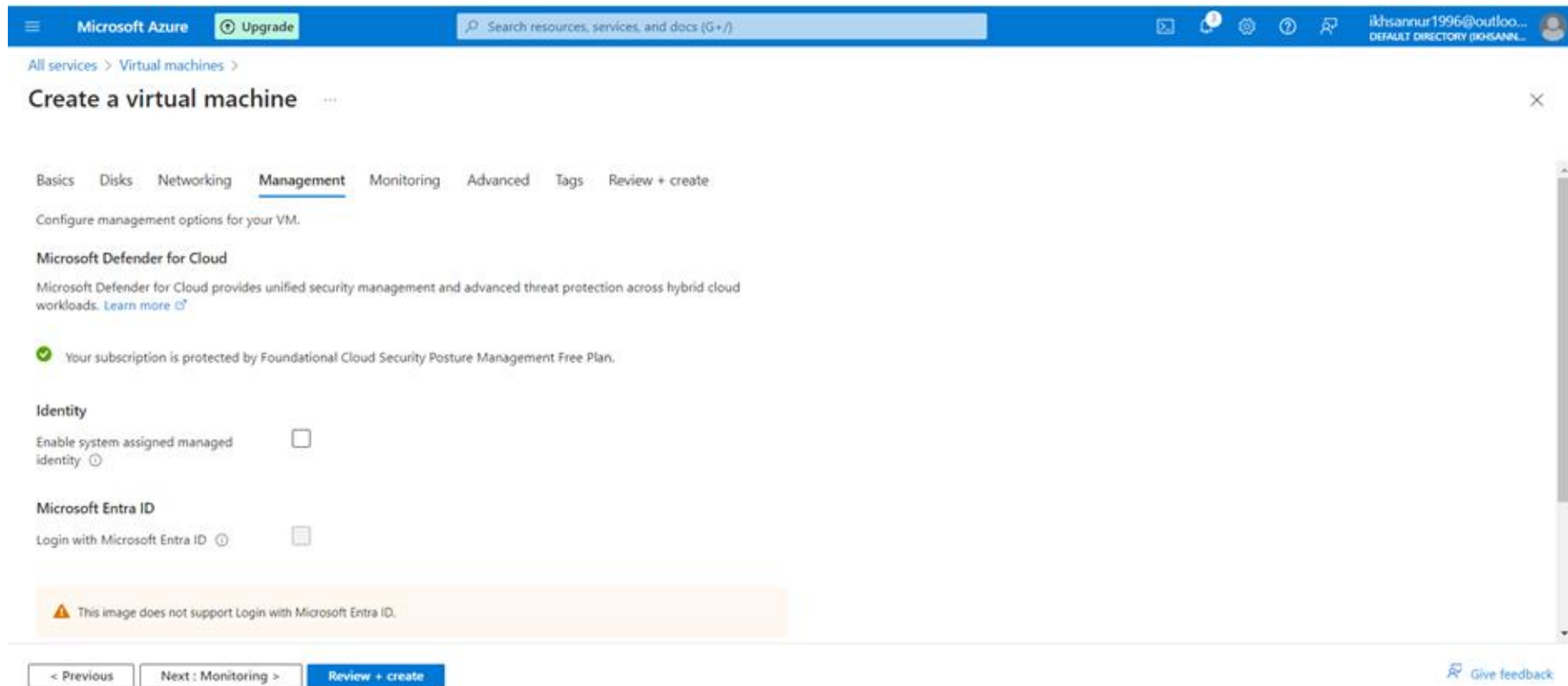
Next : Management >

Review + create

[Give feedback](#)

Select the inbound network for your virtual machines.

7. Virtual Machine Management



The screenshot shows the 'Create a virtual machine' wizard in the Microsoft Azure portal, specifically the 'Management' tab. The breadcrumb navigation at the top indicates the path: 'All services > Virtual machines > Create a virtual machine'. The 'Management' tab is selected, showing options for configuring management settings for the VM. A green checkmark indicates that the subscription is protected by the Foundational Cloud Security Posture Management Free Plan. Under the 'Identity' section, there are two checkboxes: 'Enable system assigned managed identity' and 'Login with Microsoft Entra ID', both of which are currently unchecked. A warning message at the bottom states: 'This image does not support Login with Microsoft Entra ID.' At the bottom of the page, there are three buttons: '< Previous', 'Next : Monitoring >', and 'Review + create'. A 'Give feedback' link is also present in the bottom right corner.

Microsoft Azure Upgrade Search resources, services, and docs (G+/I)

All services > Virtual machines > Create a virtual machine

Basics Disks Networking **Management** Monitoring Advanced Tags Review + create

Configure management options for your VM.

Microsoft Defender for Cloud

Microsoft Defender for Cloud provides unified security management and advanced threat protection across hybrid cloud workloads. [Learn more](#)

✓ Your subscription is protected by Foundational Cloud Security Posture Management Free Plan.

Identity

Enable system assigned managed identity ☐

Microsoft Entra ID

Login with Microsoft Entra ID ☐

⚠ This image does not support Login with Microsoft Entra ID.

< Previous Next : Monitoring > Review + create Give feedback

Configure management optional settings for your virtual machines.

8. Virtual machine Monitoring

The screenshot shows the 'Create a virtual machine' wizard in the Microsoft Azure portal, specifically the 'Monitoring' tab. The interface is in a blue header with the Microsoft Azure logo, an 'Upgrade' button, and a search bar. The user's profile is visible in the top right. The wizard has several tabs: Basics, Disks, Networking, Management, Monitoring (selected), Advanced, Tags, and Review + create. The 'Monitoring' tab contains sections for Alerts, Diagnostics, and Health, each with configuration options and checkboxes. At the bottom, there are navigation buttons: '< Previous', 'Next: Advanced >', and 'Review + create'. A 'Give feedback' link is also present in the bottom right corner.

Microsoft Azure Upgrade Search resources, services, and docs (G+/I) ikhsannur1996@outloo... DEFAULT DIRECTORY (IKHSANNUR)

All services > Virtual machines >

Create a virtual machine

Basics Disks Networking Management **Monitoring** Advanced Tags Review + create

Configure monitoring options for your VM.

Alerts

Enable recommended alert rules ☐

Diagnostics

Boot diagnostics ☒ Enable with managed storage account (recommended)
☐ Enable with custom storage account
☐ Disable

Enable OS guest diagnostics ☐

Health

Enable application health monitoring ☐

< Previous Next: Advanced > Review + create Give feedback

Monitor the performance of your virtual machines.

9. Virtual Machine Advanced Configuration

Microsoft Azure

Upgrade

Search resources, services, and docs (G+)

3

ikhsannur1996@outloo...
DEFAULT DIRECTORY (IKHSANNUR...)

All services > Virtual machines >

Create a virtual machine

BasicsDisksNetworkingManagementMonitoringAdvancedTagsReview + create

Add additional configuration, agents, scripts or applications via virtual machine extensions or cloud-init.

Extensions

Extensions provide post-deployment configuration and automation.

Extensions [Select an extension to install](#)

VM applications

VM applications contain application files that are securely and reliably downloaded on your VM after deployment. In addition to the application files, an install and uninstall script are included in the application. You can easily add or remove applications on your VM after create. [Learn more](#)

[Select a VM application to install](#)

Custom data and cloud init

Pass a cloud-init script, configuration file, or other data into the virtual machine **while it is being provisioned**. The data will be saved on the VM in a known location. [Learn more about custom data for VMs](#)

Custom data

< Previous

Next : Tags >

Review + create

[Give feedback](#)

Configure advanced settings for your virtual machines.

10. Add Tags to Virtual machines

Microsoft Azure

Upgrade

Search resources, services, and docs (G+/I)



ikhsannur1996@outloo...
DEFAULT DIRECTORY (IKHSANNUR...)

All services > Virtual machines >

Create a virtual machine

Basics Disks Networking Management Monitoring Advanced Tags Review + create

Tags are name/value pairs that enable you to categorize resources and view consolidated billing by applying the same tag to multiple resources and resource groups. [Learn more about tags](#)

Note that if you create tags and then change resource settings on other tabs, your tags will be automatically updated.

Name	Value	Resource
project	dataiku	13 selected
		13 selected

< Previous

Next : Review + create >

Review + create

Give feedback

Add tags to your virtual machines for better organization and management.

11. Review + create Virtual Machine

Microsoft Azure

Upgrade

Search resources, services, and docs (G+)

ikhsannur1996@outloo...
DEFAULT DIRECTORY (IKHSANNUR...)

All services > Virtual machines >

Create a virtual machine

Validation passed

BasicsDisksNetworkingManagementMonitoringAdvancedTagsReview + create

Price

1 X Standard E4-2as v5
by Microsoft
[Terms of use](#) | [Privacy policy](#)

Subscription credits apply ⓘ
0.2720 USD/hr
[Pricing for other VM sizes](#)

TERMS

By clicking "Create", I (a) agree to the legal terms and privacy statement(s) associated with the Marketplace offering(s) listed above; (b) authorize Microsoft to bill my current payment method for the fees associated with the offering(s), with the same billing frequency as my Azure subscription; and (c) agree that Microsoft may share my contact, usage and transactional information with the provider(s) of the offering(s) for support, billing and other transactional activities. Microsoft does not provide rights for third-party offerings. See the [Azure Marketplace Terms](#) for additional details.

NameMohamad Ikhsan Nurulloh

Preferred e-mail addressikhsannur1996@outlook.com

< PreviousNext >Create

Download a template for automationGive feedback

Review all the details and create the virtual machine.

12. Virtual Machine Success Created

The screenshot displays the Microsoft Azure portal interface for a virtual machine named 'dataiku'. The top navigation bar includes the Microsoft Azure logo, an 'Upgrade' button, a search bar, and user information for 'ikhannur1996@outloo...'. The breadcrumb trail shows the path: 'All services > CreateVm-debian.debian-11-11-gen2-20240727080242 | Overview >'. The left sidebar contains a search bar and a list of navigation options: Overview (selected), Activity log, Access control (IAM), Tags, Diagnose and solve problems, Connect, Networking, Load balancing, Application security groups, Network manager, Settings, Disks, and Extensions. The main content area features a toolbar with actions like Connect, Start, Restart, Stop, Hibernate, Capture, Delete, Refresh, Open in mobile, Feedback, and CLI / PS. Below the toolbar, the 'Essentials' section provides key information: Resource group (lkhsan), Status (Running), Location (Korea Central), Subscription (Azure subscription 1), and Subscription ID (5a3b05d8-356e-4889-9cdd-c85226ae54f2). It also lists tags (project : dataiku), operating system (Linux (debian 11)), size (Standard E4-2as v5), public IP address (20.39.202.59), virtual network/subnet (dataiku/default), DNS name (Not configured), health state (-), and time created (7/27/2024, 1:05 AM UTC). A 'JSON View' link is available. The 'Properties' tab is active, showing details for the 'Virtual machine' (Computer name: dataiku, Operating system: Linux (debian 11), VM generation: V2, VM architecture: x64, Agent status: Ready) and 'Networking' (Public IP address: 20.39.202.59, Private IP address: 10.0.0.4).

Microsoft Azure Upgrade Search resources, services, and docs (G+/)

All services > CreateVm-debian.debian-11-11-gen2-20240727080242 | Overview >

dataiku Virtual machine

Search

Connect Start Restart Stop Hibernate Capture Delete Refresh Open in mobile Feedback CLI / PS

Overview

Activity log

Access control (IAM)

Tags

Diagnose and solve problems

Connect

Connect

Bastion

Networking

Network settings

Load balancing

Application security groups

Network manager

Settings

Disks

Extensions +

Essentials

Resource group (move) : lkhsan

Status : Running

Location : Korea Central

Subscription (move) : Azure subscription 1

Subscription ID : 5a3b05d8-356e-4889-9cdd-c85226ae54f2

Tags (edit) : project : dataiku

Operating system : Linux (debian 11)

Size : Standard E4-2as v5 (2 vcpus, 32 GiB memory)

Public IP address : 20.39.202.59

Virtual network/subnet : dataiku/default

DNS name : Not configured

Health state : -

Time created : 7/27/2024, 1:05 AM UTC

JSON View

Properties Monitoring Capabilities (7) Recommendations Tutorials

Virtual machine

Computer name : dataiku

Operating system : Linux (debian 11)

VM generation : V2

VM architecture : x64

Agent status : Ready

Networking

Public IP address : 20.39.202.59 (Network interface dataiku540)

Public IP address (IPv6) : -

Private IP address : 10.0.0.4

Private IP address (IPv6) : -

Virtual network/subnet : dataiku/default

Confirm that the virtual machine has been successfully created.

Installing Dataiku

1. Installing and Configuring Dataiku

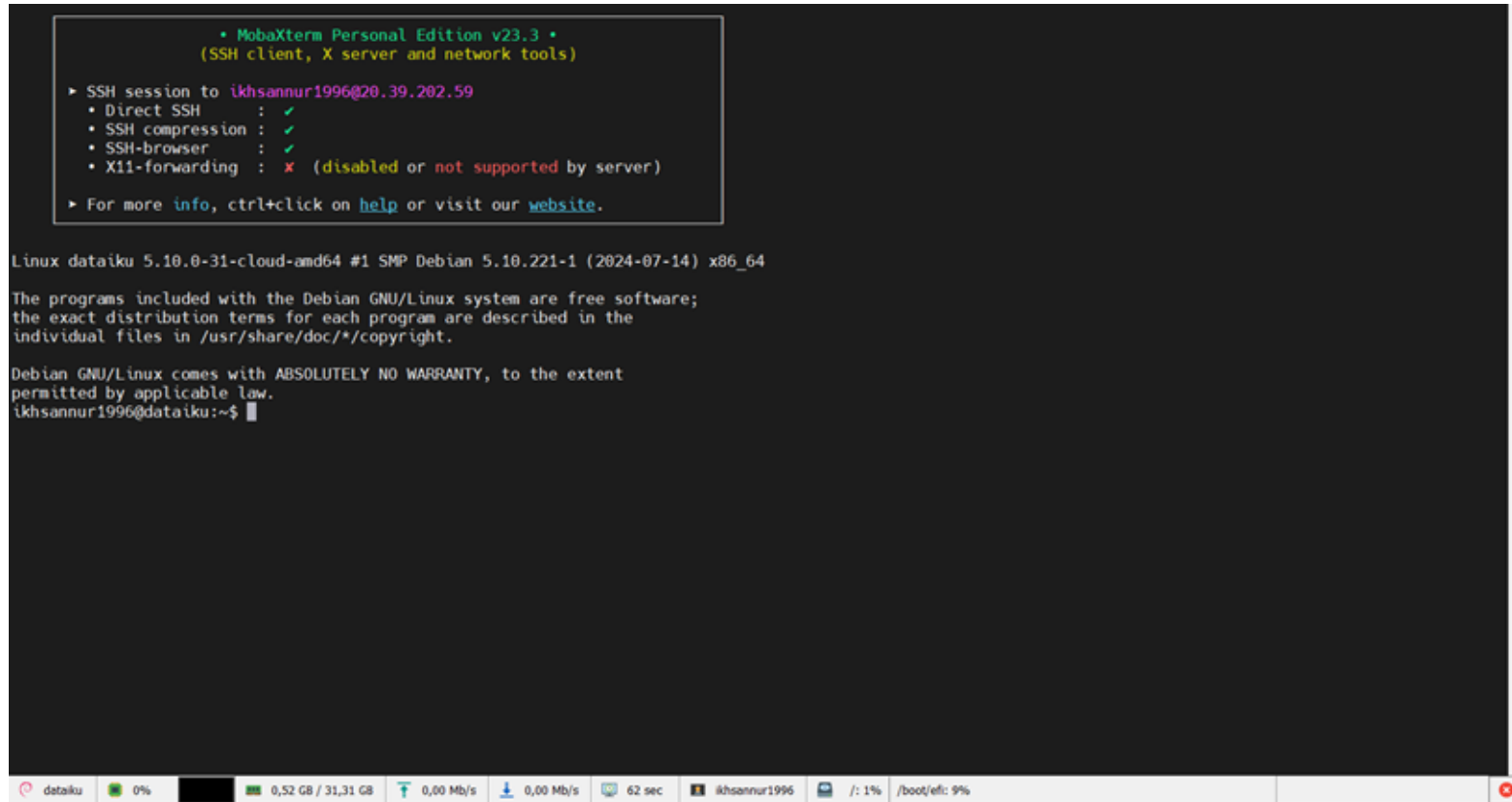
Install Dataiku on GNU/Linux

Download Dataiku and get started on Linux right away.



- **Download DSS:** Use `wget`
`https://cdn.downloads.dataiku.com/public/dss/12.6.2/dataiku-dss-12.6.2.tar.gz` or the direct link.
- **Unpack:** Extract the archive with `tar xzf dataiku-dss-12.6.2.tar.gz`.
- **Install:** Run the installation script with `dataiku-dss-12.6.2/installer.sh -d DATA_DIR -p 11000`.
- **Start:** Launch DSS with `DATA_DIR/bin/dss start`.
- **Access Studio:** Open `http://<your server address>:11000` in Chrome or Firefox.

2. SSH to the Virtual Machines



The screenshot shows a MobaXterm window with a terminal session. At the top, a title bar reads "• MobaXterm Personal Edition v23.3 • (SSH client, X server and network tools)". Below this, a status box indicates an SSH session to `ikhsannur1996@20.39.202.59`. The status details are: Direct SSH (checked), SSH compression (checked), SSH-browser (checked), and X11-forwarding (disabled or not supported by server). A link for more info is provided. The terminal output shows the Linux version: `Linux dataiku 5.10.0-31-cloud-amd64 #1 SMP Debian 5.10.221-1 (2024-07-14) x86_64`. It then displays the Debian GNU/Linux system's free software notice and warranty disclaimer. The prompt is `ikhsannur1996@dataiku:~$`. The bottom status bar shows various system metrics: dataiku, 0% CPU, 0.52 GB / 31.31 GB memory, 0.00 Mb/s network, 0.00 Mb/s disk, 62 sec session time, user ikhsannur1996, and 1% / 9% disk usage for / and /boot/efi respectively.

```
• MobaXterm Personal Edition v23.3 •
(SSH client, X server and network tools)

► SSH session to ikhsannur1996@20.39.202.59
  • Direct SSH      : ✓
  • SSH compression : ✓
  • SSH-browser     : ✓
  • X11-forwarding  : ✗ (disabled or not supported by server)

► For more info, ctrl+click on help or visit our website.

Linux dataiku 5.10.0-31-cloud-amd64 #1 SMP Debian 5.10.221-1 (2024-07-14) x86_64

The programs included with the Debian GNU/Linux system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Debian GNU/Linux comes with ABSOLUTELY NO WARRANTY, to the extent
permitted by applicable law.
ikhsannur1996@dataiku:~$
```

dataiku 0% 0,52 GB / 31,31 GB 0,00 Mb/s 0,00 Mb/s 62 sec ikhsannur1996 /: 1% /boot/efi: 9%

Use SSH to connect to the virtual machines.

3. Downloading Dataiku

```
• MobaXterm Personal Edition v23.3 •
(SSH client, X server and network tools)

► SSH session to ikhsannur1996@20.39.202.59
  • Direct SSH : ✓
  • SSH compression : ✓
  • SSH-browser : ✓
  • X11-forwarding : ✗ (disabled or not supported by server)

► For more info, ctrl+click on help or visit our website.

Linux dataiku 5.10.0-31-cloud-amd64 #1 SMP Debian 5.10.221-1 (2024-07-14) x86_64

The programs included with the Debian GNU/Linux system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Debian GNU/Linux comes with ABSOLUTELY NO WARRANTY, to the extent
permitted by applicable law.
ikhsannur1996@dataiku:~$ wget https://cdn.downloads.dataiku.com/public/dss/12.6.2/dataiku-dss-12.6.2.tar.gz
--2024-07-27 01:07:30-- https://cdn.downloads.dataiku.com/public/dss/12.6.2/dataiku-dss-12.6.2.tar.gz
Resolving cdn.downloads.dataiku.com (cdn.downloads.dataiku.com)... 18.244.61.118, 18.244.61.37, 18.244.61.64, ...
Connecting to cdn.downloads.dataiku.com (cdn.downloads.dataiku.com)|18.244.61.118|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 1812733468 (1.7G) [application/x-gzip]
Saving to: 'dataiku-dss-12.6.2.tar.gz'

dataiku-dss-12.6.2.tar.gz      52%[=====>]          ] 900.09M   346MB/s

dataiku 56% 0,54 GB / 31,31 GB 11,03 Mb/s 5994,85 Mb/s 77 sec ikhsannur1996 /: 2% /boot/efi: 9%
```

4. Update Package List and Install Required Packages



```
# Update package list  
sudo apt-get update
```

```
# Install packages  
sudo apt-get install -y acl git nginx unzip zip default-jre-headless libgomp1 python3-  
distutils
```

Ensure your system is ready for Dataiku installation by updating the package list and installing the required packages.

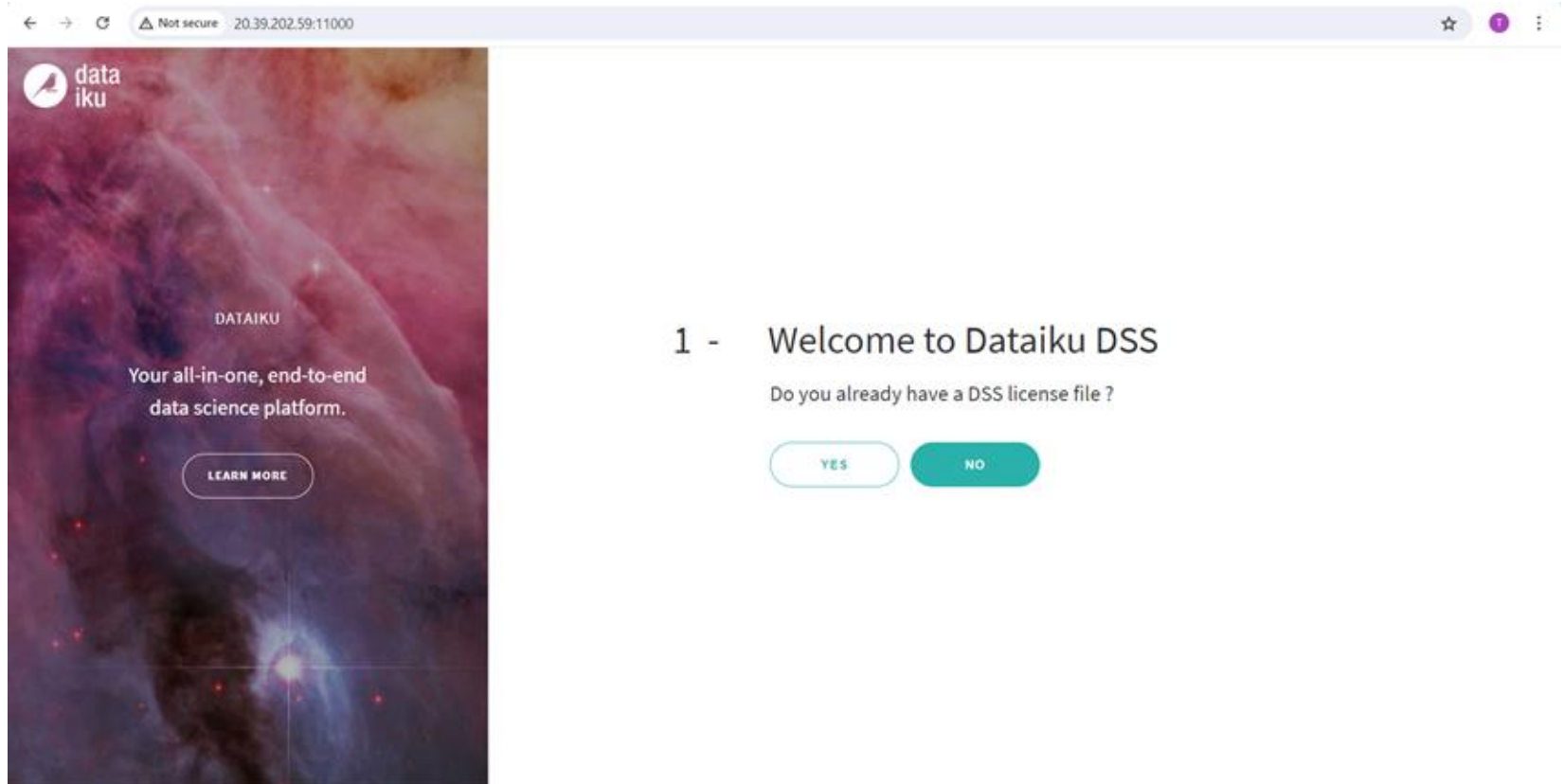
5. Installing Dataiku

```
/home/ikhsannur1996/dataiku-dss-12.6.2/scripts/_startup.inc.sh: line 236: warning: setlocale: LC_ALL: cannot change locale (en_US.UTF-8): No such file or directory
/bin/bash: warning: setlocale: LC_ALL: cannot change locale (en_US.UTF-8)
Extension jupyter-js-widgets/extension enabled successfully
/home/ikhsannur1996/dataiku-dss-12.6.2/scripts/_startup.inc.sh: line 236: warning: setlocale: LC_ALL: cannot change locale (en_US.UTF-8): No such file or directory
[+] Generating default env file
/bin/bash: warning: setlocale: LC_ALL: cannot change locale (en_US.UTF-8)
[+] Generating supervisor configuration
/bin/bash: warning: setlocale: LC_ALL: cannot change locale (en_US.UTF-8)
[+] Generating nginx configuration
/bin/bash: warning: setlocale: LC_ALL: cannot change locale (en_US.UTF-8)
/bin/bash: warning: setlocale: LC_ALL: cannot change locale (en_US.UTF-8)
*****
* Installation complete (DSS node type: design)
* Next, start DSS using:
*   '/home/ikhsannur1996/DATA_DIR/bin/dss start'
* Dataiku DSS will be accessible on http://<SERVER ADDRESS>:11000
*
* You can configure Dataiku DSS to start automatically at server boot with:
*   sudo -i "/home/ikhsannur1996/dataiku-dss-12.6.2/scripts/install/install-boot.sh" "/home/ikhsannur1996/DATA_DIR" ikhsannur1996
*****
ikhsannur1996@dataiku:~$ /home/ikhsannur1996/DATA_DIR/bin/dss start
/home/ikhsannur1996/dataiku-dss-12.6.2/scripts/_startup.inc.sh: line 236: warning: setlocale: LC_ALL: cannot change locale (en_US.UTF-8): No such file or directory
/bin/bash: warning: setlocale: LC_ALL: cannot change locale (en_US.UTF-8)
/bin/bash: warning: setlocale: LC_ALL: cannot change locale (en_US.UTF-8)
/bin/bash: warning: setlocale: LC_ALL: cannot change locale (en_US.UTF-8)
/bin/bash: warning: setlocale: LC_ALL: cannot change locale (en_US.UTF-8)
Waiting for DSS supervisor to start ...
/bin/bash: warning: setlocale: LC_ALL: cannot change locale (en_US.UTF-8)
/bin/bash: warning: setlocale: LC_ALL: cannot change locale (en_US.UTF-8)
backend          STARTING
ipython          STARTING
nginx            STARTING
DSS started, pid=6241
/bin/bash: warning: setlocale: LC_ALL: cannot change locale (en_US.UTF-8)
/bin/bash: warning: setlocale: LC_ALL: cannot change locale (en_US.UTF-8)
/bin/bash: warning: setlocale: LC_ALL: cannot change locale (en_US.UTF-8)
Waiting for DSS backend to start ....
```

dataiku 98% 1,62 GB / 31,31 GB 0,10 Mb/s 9,83 Mb/s 305 sec ikhsannur1996 /: 6% /boot/efi: 9%

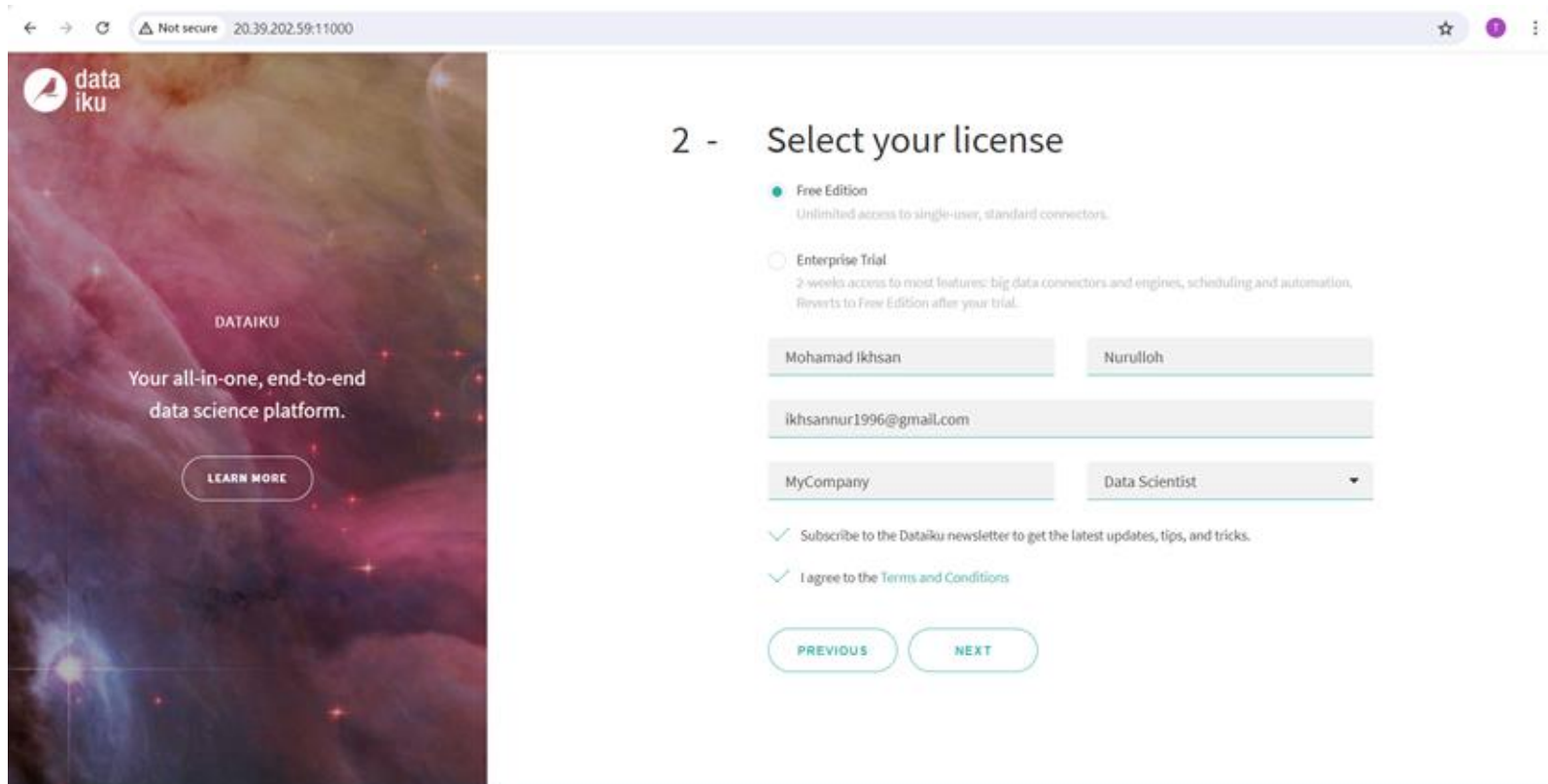
Run the installation script with `dataiku-dss-12.6.2/installer.sh -d DATA\_DIR -p 11000`.

6. Welcome Page Configuration Select the license file



Set up the welcome page settings for Dataiku. Choose 'No' if you do not possess a DSS license.

7. Select Dataiku License



← → ↺ Not secure 20.39.202.59:11000 ☆ ⓘ ⋮



data
iku

DATAIKU

Your all-in-one, end-to-end
data science platform.

LEARN MORE

2 - Select your license

- ☒ Free Edition
Unlimited access to single-user, standard connectors.
- ☐ Enterprise Trial
2 weeks access to most features: big data connectors and engines, scheduling and automation.
Reverts to Free Edition after your trial.

Mohamad Ikhsan Nurulloh

ikhsannur1996@gmail.com

MyCompany Data Scientist ▼

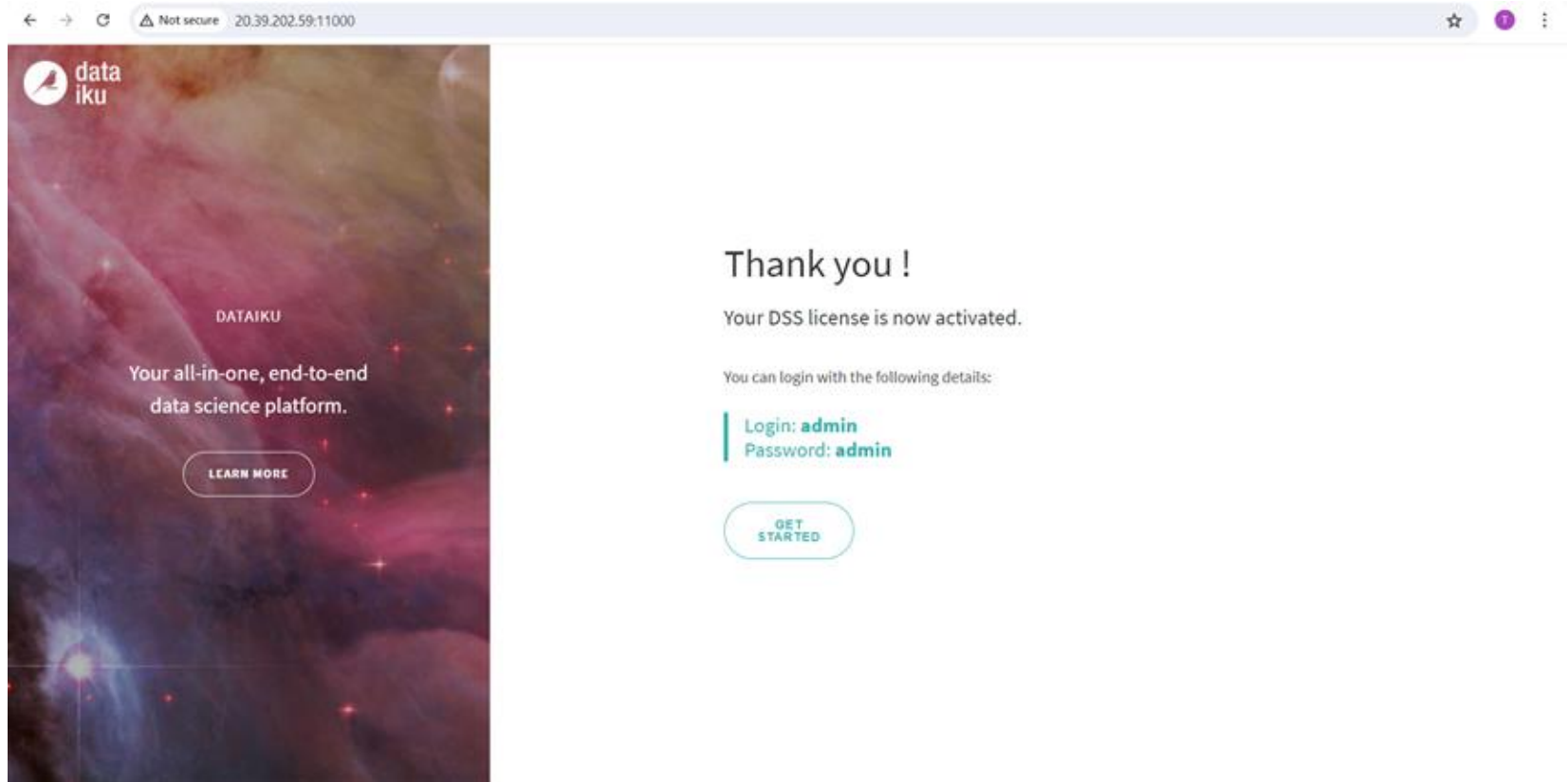
✓ Subscribe to the Dataiku newsletter to get the latest updates, tips, and tricks.

✓ I agree to the [Terms and Conditions](#)

PREVIOUS NEXT

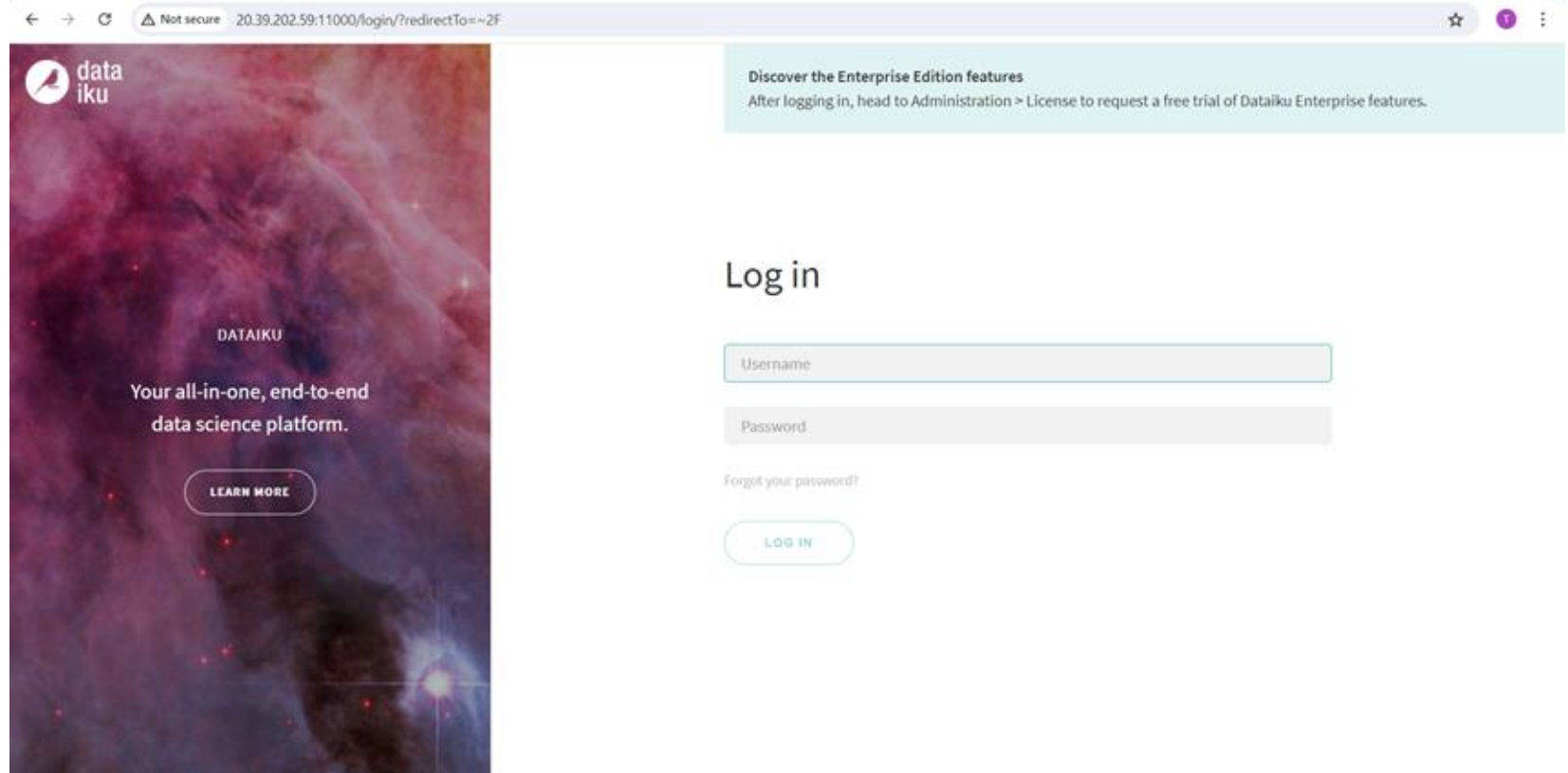
Select the license file for Dataiku: Free Edition or Enterprise Trial.

8. Login and Password Generated



Generate login credentials for Dataiku.

9. Login Page Dataiku



The screenshot shows the Dataiku login page in a web browser. The browser's address bar displays the URL `20.39.202.59:11000/login/?redirectTo=~2F` and indicates the connection is "Not secure". The page features a large, vibrant nebula background image. On the left side, the Dataiku logo is positioned at the top, followed by the text "DATAIKU" and "Your all-in-one, end-to-end data science platform." Below this is a "LEARN MORE" button. On the right side, there is a light blue banner with the text "Discover the Enterprise Edition features" and a link to "Administration > License". Below the banner, the "Log in" section contains two input fields for "Username" and "Password", a "Forgot your password?" link, and a "LOG IN" button.

dataiku

DATAIKU

Your all-in-one, end-to-end data science platform.

LEARN MORE

Discover the Enterprise Edition features

After logging in, head to Administration > License to request a free trial of Dataiku Enterprise features.

Log in

Username

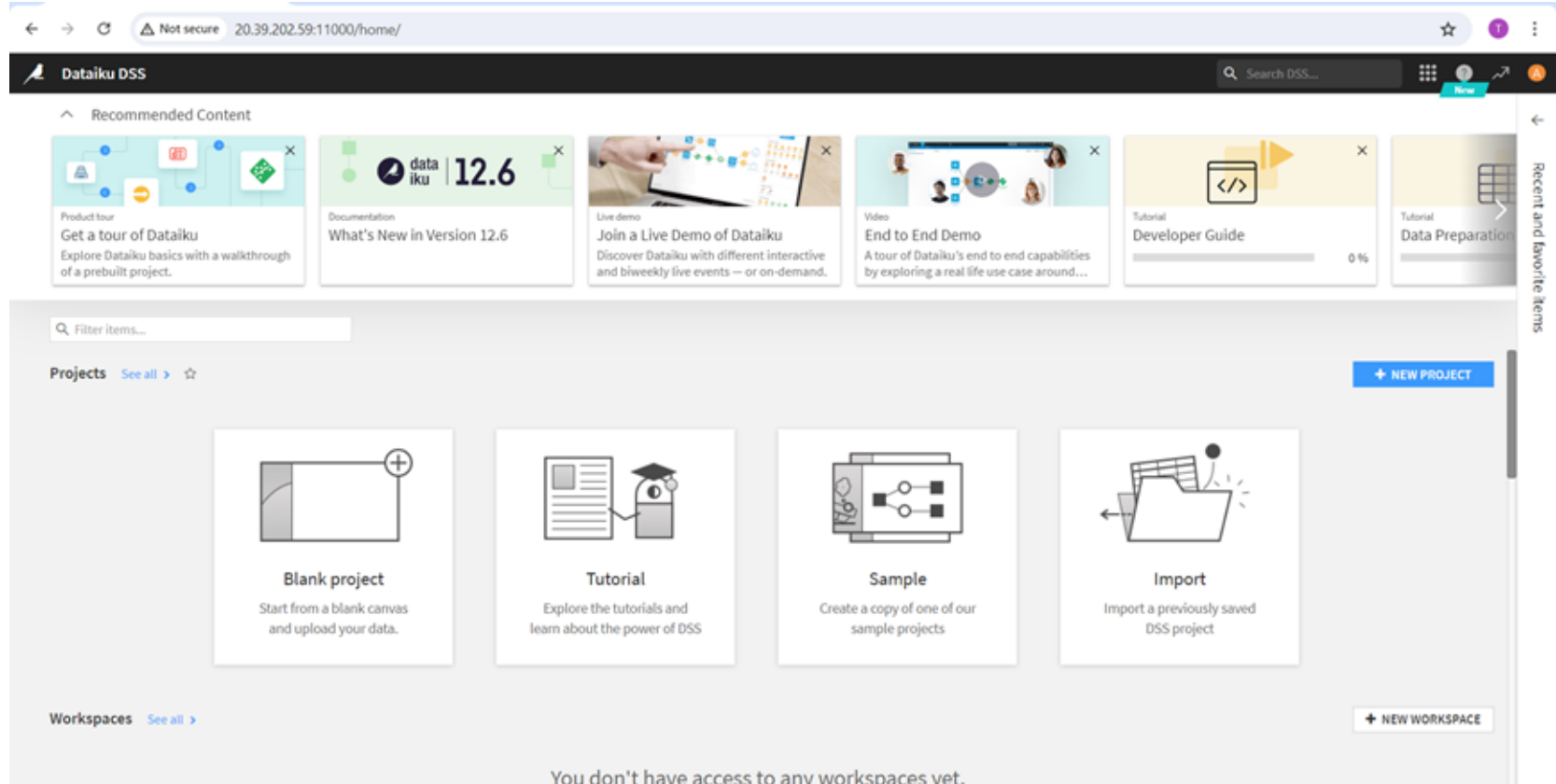
Password

[Forgot your password?](#)

LOG IN

Log in to the Dataiku platform.

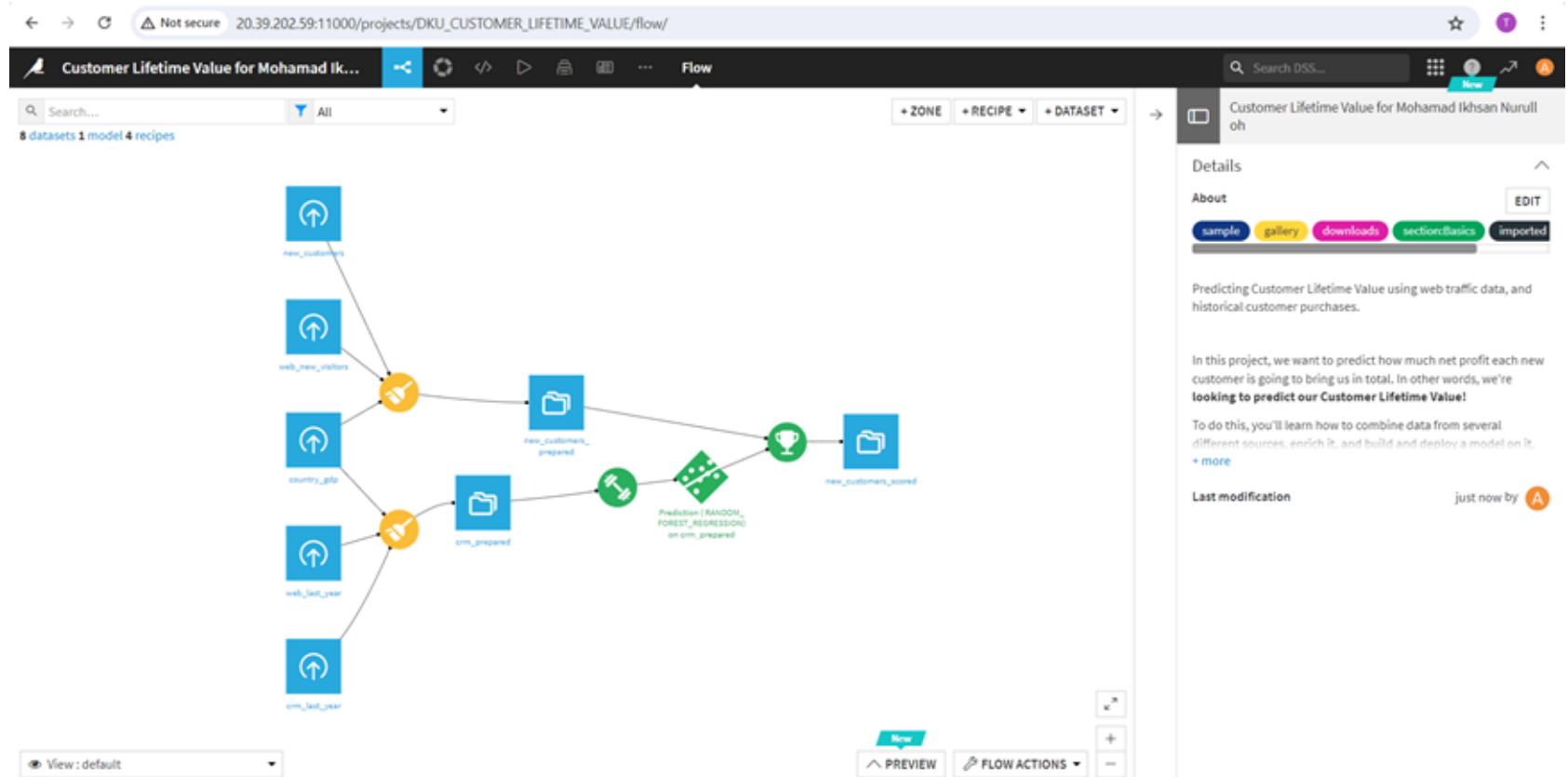
10. Dataiku Home Page



The screenshot shows the Dataiku DSS home page in a web browser. The browser's address bar displays "20.39.202.59:11000/home/" with a "Not secure" warning. The page header includes the "Dataiku DSS" logo, a search bar labeled "Search DSS...", and user profile icons. Below the header, a "Recommended Content" section features six tiles: "Product tour" (Get a tour of Dataiku), "Documentation" (What's New in Version 12.6), "Live demo" (Join a Live Demo of Dataiku), "Video" (End to End Demo), "Tutorial" (Developer Guide), and another "Tutorial" (Data Preparation). The main content area has a "Filter items..." search bar and a "Projects" section with a "See all" link and a star icon. To the right of the projects is a blue button labeled "+ NEW PROJECT". Below the projects are four large cards: "Blank project" (Start from a blank canvas and upload your data.), "Tutorial" (Explore the tutorials and learn about the power of DSS), "Sample" (Create a copy of one of our sample projects), and "Import" (Import a previously saved DSS project). At the bottom left, there is a "Workspaces" section with a "See all" link. At the bottom right, there is a button labeled "+ NEW WORKSPACE". A message at the very bottom states: "You don't have access to any workspaces yet."

Access the Dataiku home page.

11. Sample Dataiku Project



Open a sample project in Dataiku.

Testing PostgreSQL Connection

1. Testing Connection To PostgreSQL

The screenshot shows the 'New PostgreSQL connection' configuration page in Dataiku DSS. The browser address bar shows the URL: 20.39.202.59:11000/admin/connections/new/PostgreSQL/. The page has a dark header with 'Dataiku DSS' and a search bar. Below the header is a navigation bar with tabs: DSS settings, License, Connections, Security, Clusters, Code Studios, Code Envs, Monitoring, Maintenance, and Settings. The 'Connections' tab is active.

The main content area is titled 'New PostgreSQL connection'. It includes a form for creating a new connection. The 'New connection name' field is filled with 'dataiku-dataset'. Below this are sections for 'Basic params', 'Advanced JDBC properties', and 'Advanced params'. The 'Basic params' section includes fields for Host, Database, Port, User, and Password. The 'Advanced JDBC properties' section includes a checkbox for 'Use custom JDBC URL' and a '+ ADD PROPERTY' button. The 'Advanced params' section includes a checkbox for 'Allow schema override' and fields for 'Table prefix' and 'Table suffix'. The 'Naming rules for new datasets' section explains that these settings define how managed SQL datasets are mapped to database tables.

At the bottom of the page, there is a green bar with the text 'Connection OK'. Below this bar is a link 'Import tables to datasets'.

Test the connection to the PostgreSQL database from Dataiku.

2. Testing Connection To PostgreSQL

The screenshot shows the 'DSS settings' page for a PostgreSQL connection in Dataiku DSS. The interface is divided into several sections:

- to Spark**: Includes an experimental checkbox for adding driver jars to spark-submit subprocesses.
- path**: A text input field for the search path, with a note that it will be set on all sessions.
- mmmands**: A list of statements to run after the connection is created.
- ct commands**: A list of additional statements to run after the connection is created when used in a job context.
- ir data**: A checkbox for truncating tables instead of dropping them when columns haven't changed.
- if you choose to index this connection to make it searchable in the DSS data catalog**: Two checkboxes for indexing database indices and foreign key relationships.
- ys**: A checkbox for fetching and indexing foreign key relationships.
- Date settings**: A dropdown for the default assumed timezone, currently set to 'Local time zone'.
- Custom properties**: A table of advanced connection properties, including 'ssl' (set to 'true') and 'ssifactory' (set to 'org.postgresql.LazyNonValidatingFact').
- Usage params**: Checkboxes for 'Allow write' and 'Allow managed datasets', both of which are checked.
- Max nb. of activities**: A text input field for limiting the number of concurrent activities.
- Security settings**: A section at the bottom of the settings page.

At the bottom of the interface, there are buttons for 'TEST' and 'CREATE', and a link to 'Import tables to datasets'.

If PostgreSQL doesn't use SSH, you should configure custom properties with `ssh=true`.

References

<https://doc.dataiku.com/dss/latest/>

<https://doc.dataiku.com/dss/latest/installation/custom/initial-install.html>

<https://www.dataiku.com/product/get-started/linux/>

<https://doc.dataiku.com/dss/latest/connecting/sql/postgresql.html>

<https://learn.microsoft.com/en-us/azure/postgresql/>

<https://learn.microsoft.com/en-us/azure/virtual-machines/>

<https://learn.microsoft.com/en-us/azure/virtual-network/>

Thank You